

Interpretable Authorship Attribution of Five Frontier LLMs Across Seven Languages

Adrian Erlikhman

High School Student

Los Angeles Center for Enriched Studies Los Angeles Center for Enriched Studies

Los Angeles, CA, USA

erlikhman.adrian@gmail.com

Michael Tarekegn

High School Student

Los Angeles, CA, USA

Philo Juang

University of California, Los Angeles

Los Angeles, CA, USA

Abstract—Classifiers that attribute machine-generated text to the model that wrote it are built and evaluated mostly in English, and the cost of an interpretable representation against character n -grams has not been measured on content-controlled multilingual data. We collected 840 persuasive essays from five frontier large language models (GPT-5.5, Gemini 3.5 Flash, Claude Opus 4.7, Grok 4.3, DeepSeek V4 Pro) in seven languages from 12 content-matched prompts, described each text by 21 features computed the same way on Universal Dependencies parses in every language, and scored five-way authorship attribution with leave-one-prompt-out cross-validation. Logistic regression reaches .59–.72 accuracy in every language (chance 20%). Across a coarse pre-registered resource ranking we detect no decline (-0.038 log-odds per rank step, 95% CI -0.12 to +0.04), an interval that admits a drop of up to 17 points. Character n -grams are 3–22 points more accurate within a language (Holm-significant in 2 of seven); given the same unlabelled adaptation to the target language, the features transfer between languages at mean .49 against .41 for n -grams. In each language, five features chosen on the training folds reach 78–106% of the full set’s macro-F1. Model attribution on machine-translated text stays at .57–.70, and the five models, asked which model wrote each text, score 20.3%–25.4%.

Index Terms—authorship attribution, AI-generated text, multilingual text mining, interpretable features, large language models

1. Introduction

Most work on machine-generated text asks whether a text was written by a person or by a model. A second question matters for provenance: *which* model wrote it, which bears on tracing influence campaigns and enforcing provider-specific rules. The model-attribution literature is mostly English [1], [2], and the two multilingual studies we know of use fine-tuned or probability-based classifiers whose decisions cannot be read [3], [4].

There are two reasons to doubt that English results carry over. Models see far less training text in lower-resource

languages [5], so their writing there may be less distinctive, and languages differ in script and morphology, so many stylometric measures are undefined outside English while those that are defined may track the language more than the author. Character n -grams, the single most successful feature in authorship attribution [6], have the opposite problem: they see vocabulary and morphology, so they are tied to the script and lexicon they were fit on.

We address both problems with a design that holds content fixed across languages and describes every text with features computed the same way on Universal Dependencies (UD) parses [7], and we measure its cost against an n -gram baseline. Six questions were written down before data collection: whether five frontier models can be told apart within each of seven languages (**RQ1**), whether model attribution weakens as a language’s resource level falls (**RQ2**), whether a model’s style is shared across languages or specific to each (**RQ3**), whether the models converge on one style in lower-resource languages (**RQ4**), whether machine translation erases the signal (**RQ5**) and, as a baseline, whether the models themselves can identify the author (**RQ6**). The n -gram baseline and a feature-economy analysis were added afterwards.

Our contributions are (i) a content-controlled corpus of 840 essays (5 models $\times$ 7 languages $\times$ 12 prompts $\times$ 2 generations) with 1,440 machine translations, and a 21-feature representation with one computational definition in every language that reaches .59–.72 five-way accuracy in all seven languages; (ii) two baselines that bracket it: character n -grams, 3–22 points better within a language and worse across languages under equal unlabelled adaptation (.41 against the features’ .49), and the models themselves at 20.3%–25.4%, as Bai et al. [8] found in English; (iii) an account of which features carry the signal and which survive machine translation. Two pre-registered questions (resource gradient, convergence) return nulls.

2. Related Work

Detection and stylometry. Human-versus-machine detectors fine-tuned on English lose about a quarter of their

F1 on other languages [9]. For human authors, character n -grams are the single most successful feature [6], masking all but the most frequent words isolates a structural signal [10], and UD-derived syntactic features attribute human authors in several languages [11].

Model attribution. In English, Uchendu et al. framed authorship attribution among neural generators [1], Kumarage and Liu applied stylometric features to LLMs [12], Sun et al. showed that embedding classifiers separate model families and largely survive LLM translation into Chinese [2], and models optimized to fool detectors keep a distinct style [13]. La Cava et al. attribute among seven older LLMs and human text in 18 languages with fine-tuned and probability-based detectors and find generators harder to distinguish in English [3]; MultiGhostBench attributes long-form text to five recent LLMs in six languages and finds that fine-tuned transformers retain generator information across languages while statistical detectors are more language-dependent [4]. In Japanese, Zaitzu et al. attribute among six LLMs with part-of-speech and function-word features [14]. To our knowledge, no prior which-model study uses one feature set with the same computational definition in every language, holds content fixed across languages, and compares that representation with character n -grams under matched adaptation within and across languages.

Homogenization and LLM judges. Writing with LLMs reduces the diversity of human text [15]; whether models converge on each other as training data thins has not been tested, and the one multilingual observation points the other way [3]. LLM evaluators favour their own outputs [16], and models asked to name the author of a single English text score near chance and default to GPT or Claude [8]. RQ6 repeats that task in seven languages; a companion study examines self-recognition in English [17].

3. Data and Method

3.1. Models, languages and prompts

The five subject models (GPT-5.5, Gemini 3.5 Flash, Claude Opus 4.7, Grok 4.3, DeepSeek V4 Pro) were queried through OpenRouter in September 2026, with the served model identifier logged for every response.¹ The seven languages are English, Spanish, Chinese, Russian, Japanese, Turkish and Hindi, assigned ordinal resource ranks 1–7 in that order before data collection as a coarse proxy for their share of web training data. The ranking is contestable: Joshi et al. [5] place English, Spanish, Chinese and Japanese in their highest class (5) and Russian, Turkish and Hindi one class lower (4), and corpus-based orderings disagree with each other, so we report the two-class split beside the pre-registered order.

Twelve English prompt schemas each fix a topic, a stance (six for, six against), three supporting claims, a prose-only instruction and a reading-level tier. For each language

a model that is not a subject (Llama 4 Maverick) wrote a native prompt from the schema rather than translating the English, so the wording is idiomatic while the content is fixed. A model from a third family (Qwen 3.7 Plus) read each of the 84 prompts blind, extracted topic, stance and claims, and compared them with the schema; failed prompts were regenerated with the reviewer’s note until all 84 passed.

3.2. Generation, validation and features

Each prompt was sent as a single user turn with no system prompt at temperature 0.7, twice per cell with fixed seeds and hidden reasoning excluded from the output, giving $5 \times 12 \times 7 \times 2 = 840$ responses. Each passed language identification (confidence ≥ 0.6), a 150–900 word-equivalent window and a per-language refusal check. One Grok response to a Hindi prompt was written in English and excluded, leaving 839 texts; ten DeepSeek responses that exhausted their token budget on hidden reasoning were re-collected with a larger budget.

Every text was parsed with Stanza [18], with Chinese and Japanese pre-split on sentence-final punctuation. We compute 21 features in five groups. *Lexical*: moving-average type–token ratio (MATTR, window 50), hapax rate, mean token length, Zipf slope. *Syntactic*: mean and SD of sentence length, burstiness of sentence length, mean maximum dependency depth, subordinate-clause rate (acl, advcl, ccomp, xcomp, csubj per sentence), function-word ratio, first-person rate. *Structure*: paragraph count and length, question rate, connective rate. *Punctuation*: commas, colons, dashes and semicolons per 1,000 tokens, with full-width and script variants mapped to the same feature. *Character*: character-bigram entropy and digit rate. Eight features depend on the parser or sentence splitter; the other thirteen need only tokenization.

3.3. Analysis

The two generations of a prompt are not independent, so every cross-validation split is leave-one-prompt-out (LOPO, 12 folds), and accuracy intervals and GLM standard errors are prompt-clustered, while binomial, exact McNemar and judge tests treat texts as independent. The 28 headline comparisons (features and n -grams vs. chance, features vs. n -grams, features vs. the best judge, in seven languages) are Holm-corrected as one family; Spearman p -values over seven languages are exact permutation values. The pre-registered plan is docs/analysis_plan.md in the repository.

RQ1. Within each language, multinomial logistic regression on standardized features. **Baseline.** TF-IDF over character 1–4-grams fit inside each training fold, then logistic regression; word 1–2-grams likewise. **RQ2.** A binomial GLM of cell-level correctness on resource rank with clustered standard errors (by prompt, by language, and by both), also with features residualized on log length within each language. **RQ3.** For each feature, a two-way ANOVA (model $\times$ language, prompt as a blocking factor) with partial η^2 ; and

1. OpenRouter identifiers: openai/gpt-5.5, google/gemini-3.5-flash, anthropic/claude-opus-4.7, x-ai/grok-4.3, deepseek/deepseek-v4-pro-0813.

TABLE 1. PER-LANGUAGE RESULTS IN RESOURCE-RANK ORDER. UD: LOGISTIC REGRESSION ON THE 21 FEATURES, ACCURACY [PROMPT-CLUSTERED 95% CI] AND MACRO-F1; n -GRAM: CHARACTER 1–4-GRAM TF-IDF [CI]; JUDGES: MEAN OF THE FIVE LLM JUDGES (BEST); SEP.: MEAN CENTROID DISTANCE (z -UNITS). CHANCE IS 0.20.

Language	UD [95% CI]	F1	n -gram [CI]	Judges	Sep.
English	.717 [.61–.82]	.711	.750 [.71–.79]	.230 (.267)	3.59
Spanish	.625 [.53–.70]	.625	.733 [.65–.80]	.207 (.217)	3.11
Chinese	.650 [.55–.73]	.644	.867 [.81–.93]	.208 (.233)	3.01
Russian	.650 [.52–.78]	.654	.783 [.66–.88]	.237 (.258)	2.97
Japanese	.708 [.63–.78]	.707	.792 [.71–.87]	.198 (.225)	2.70
Turkish	.592 [.50–.69]	.596	.783 [.73–.84]	.205 (.258)	2.84
Hindi	.639 [.56–.71]	.634	.782 [.73–.83]	.247 (.361)	2.69

a transfer matrix in which features are z -scored within each language and a classifier trained on language A is tested on B . For n -grams the counterpart fits the vocabulary on the union of A and B without labels and z -scores TF-IDF columns within each language, so both representations receive the same unlabelled adaptation. **RQ4.** After z -scoring within each language, the mean pairwise distance between the five model centroids, the silhouette of model labels and the between/within scatter ratio, with stratified bootstrap intervals. **RQ5.** All 120 English responses were translated into the six other languages by Google Translate and by an open model that is not a subject (MiniMax M3); we measure model attribution on the translations alone, transfer from classifiers trained on the English originals or on native target-language responses (features z -scored within each set), and per-feature Spearman correlations between originals and translations. **RQ6.** Each subject model was asked, one text at a time with a shuffled candidate list and no metadata, which of the five models wrote it (temperature 0, hidden reasoning disabled where the API allows it), following the exact-model-prediction task of Bai et al. [8]. **Feature economy.** A nested feature-selection curve (within each LOPO fold, features are ranked by mean absolute standardized coefficient on the training fold only and the top k are refit, for $k = 1, \dots, 21$); a per-language ablation of each feature group; and a character model that sees only punctuation, digits and whitespace, every word replaced by a placeholder [10], scored LOPO within language.

4. Results

4.1. Model attribution works in every language, at a price

Table 1 shows five-way accuracy of .59–.72 in every language; every prompt-clustered interval excludes chance and every Holm-corrected binomial has $p < 10^{-4}$. Length is part of the signal: with features residualized on log length, logistic regression falls to .44–.57, still far above chance. Dropping the whole structure group costs at most .07 macro-F1 (Sec. 4.2), so the signal is not paragraph count alone.

The baseline is more accurate within a language. Character n -grams reach .73–.87, 3–22 points above the features (mean 13). Per-language McNemar tests on about

120 paired predictions are underpowered for gaps near ten points (Spanish: 11 points, Holm $p = 0.198$); the advantage is Holm-significant in Chinese, Turkish and in the same direction everywhere else. This is the expected result [6]: n -grams see vocabulary and morphology, and the features do not.

LLM judges (RQ6). Asked one text at a time which of the five models wrote it, the judges score 20.3% (DeepSeek) to 25.4% (Claude). Only Claude is above chance ($p < 10^{-4}$), and only Claude names its own text more often than others’ (own-text recall 17% against 7% false self-claims; Fisher $p < 10^{-4}$), the registered prediction; GPT-5.5 names itself for 82% of texts regardless of author. The feature classifier exceeds the best judge by 28–48 points in every language (Holm $p < 10^{-4}$). Bai et al. [8] report the same near-chance, GPT- and Claude-biased behaviour in English; it holds in all seven languages here. A companion study shows that a side-by-side lineup with reasoning enabled raises the same judges [17].

4.2. What carries the signal

By mean absolute standardized coefficient the strongest features are paragraph count, comma rate, char-bigram entropy, paragraph length, Zipf slope. In the two-way ANOVA, language explains far more feature variance than model (mean partial η^2 .54 against .145; interaction .13; prompt .16) and dominates every measure tied to morphology or vocabulary, while model effects are largest for hapax rate (.38), Zipf slope (.37), comma rate (.32), char-bigram entropy (.31), paragraph count (.29) and smallest for subordination, question rate and dependency depth ($\leq .03$): the models differ in lexical variety, punctuation and paragraphing, and hardly at all in clause structure.

The nested selection curve (Fig. 1a) ranks features on the training folds only. In each language the five highest-ranked features reach 78–106% of the full set’s macro-F1 (.50–.75 against .60–.71), but the smallest number of features within 0.03 of a language’s best score ranges from 4 (Russian) to 13 (Turkish), and the ranking changes with language: paragraph count is first in Russian and sixteenth in English. The pooled order is paragraph count, paragraph length, sentence length, comma rate, hapax rate. No single group is required: dropping any one costs at most .11 macro-F1 (punctuation in Hindi, syntax in Turkish); structure alone gives .31–.61 and punctuation alone .26–.53. The punctuation, digit and whitespace model attributes at .49–.81 within language, so segmentation, punctuation and length alone carry a signal.

4.3. Across languages and through translation

Cross-language transfer (RQ3). After within-language z -scoring (Fig. 2a), a classifier trained on any language reaches .37–.59 on any other (mean .49, against .65 within language), all 42 off-diagonal cells exceed chance after Holm correction, and 63% of the above-chance accuracy transfers. Given the same unlabelled adaptation, character n -grams transfer at mean .41 (Fig. 2b), above chance in 41

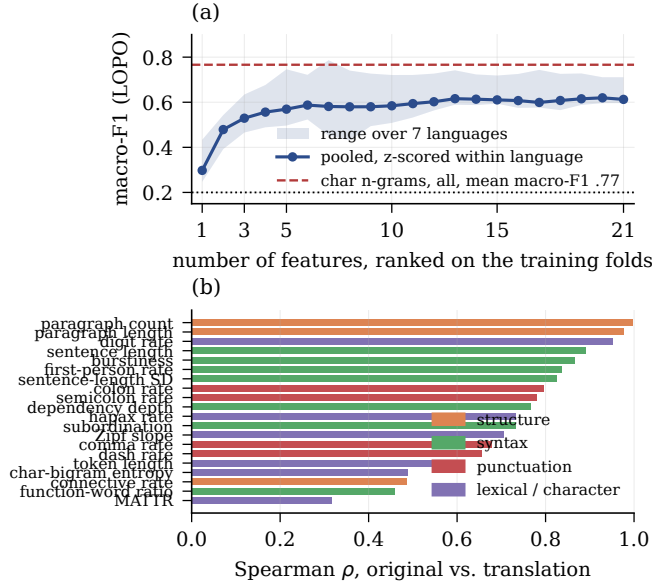


Figure 1. (a) Nested feature-selection curve; features are ranked on the training folds only, and the band is the range over the seven per-language curves. (b) Spearman correlation between each English original and its machine translation, per feature, averaged over six languages and two translators; question rate is omitted because the English essays contain no questions.

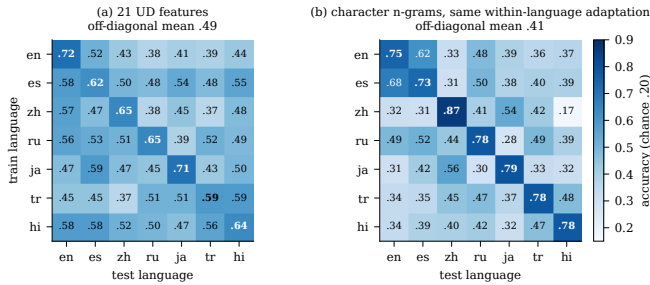


Figure 2. Cross-language transfer: a classifier trained on one language (rows) is tested on another (columns) after within-language z-scoring; the diagonal is LOPO within language. (a) UD features. (b) Character n -grams given the same adaptation (union vocabulary, TF-IDF columns z-scored per language).

of 42 cells; the features are better in 33 of the 42 cells, and n -grams keep their edge where script and vocabulary are shared. Without adaptation, n -grams transfer at .28 with 14 constant-prediction cells. Two caveats: the test language contains the 12 training prompts, so off-diagonal cells are an upper bound for unseen prompts, and both representations use the test language’s own unlabelled statistics.

Translation (RQ5). Classifiers trained and tested on the translations alone (LOPO) reach .57–.70, every interval excluding chance, about 9 points below the English originals (.717) and close to native responses in the six languages (mean .644); n -grams retrained on the translations reach .63–.78. Which features pass through (Fig. 1b): paragraph count ($\rho = 1.00$), paragraph length (.98), digit rate (.95) and sentence length (.89) are preserved largely by construc-

tion, since both translators keep paragraph and sentence boundaries and copy digits; hapax rate (.73), Zipf slope (.70) and comma rate (.68) survive with some loss; MATTR (.32), function-word ratio (.46) and bigram entropy (.49) are rewritten by the target language. A classifier trained on the English originals is correct on .66 of their translations with either translator; because it has seen every source text, this is a same-text stability measure rather than accuracy on new texts. A classifier trained on native responses in the target language, which has seen none of these texts, reads them at .521–.531. An English-trained n -gram model with no vocabulary adaptation scores .20–.28 on every target language except Spanish (.57), including Latin-script Turkish, so the loss is vocabulary rather than script; the adapted counterpart was not run on translations.

4.4. Two pre-registered questions with null answers

Resource gradient (RQ2). The cell-level GLM gives $\beta = -0.038$ log-odds per rank step (SE 0.042, 95% CI -0.12 to +0.04, $p = 0.361$ clustered by prompt; $p = 0.226$ by language, a standard error resting on seven clusters; $p = 0.343$ by both). The point estimate is under one accuracy point per step, and the ordering is not monotone (Japanese matches English; Hindi exceeds Spanish and Turkish), all within intervals of about ± 9 points. The minimum detectable effect at 80% power is about 16 points over the six steps and the interval admits a decline of up to 17, so this is a null with bounded power rather than a demonstration of invariance. Joshi’s two classes give the same picture (mean .675 for the four Class-5 languages against .627 for the three Class-4 ones), as does the n -gram baseline (Hindi .782 exceeds English .750; GLM $p = 0.380$). The pre-specified criterion (negative coefficient, $p < 0.05$) is not met.

Convergence (RQ4). Mean distance between model centroids falls from 3.59 z -units in English to 2.69 in Hindi (Spearman $\rho = -0.96$ against rank, $p = 0.003$) and the between/within scatter ratio from .41 to .18 ($\rho = -1.00$, $p < 0.001$), with the Hindi intervals below the English ones; but the silhouette is -0.02 to 0.04 in every language, its English and Hindi intervals overlap ([-0.02, 0.03] against [-0.08, -0.02]), and the pre-registered rule (all three intervals separated) is not met. The change is located in one model: the mean of the four pairwise distances involving Grok falls from 4.88 in English to 2.78 in Hindi, while the mean of the six other pairs is 2.73 in English and 2.63 in Hindi ($\rho = -0.46$, $p = 0.302$). Grok’s English register is the outlier that disappears outside English; the other four models are as far apart in Hindi as in English, and silhouettes near zero mean there is no cluster structure in the full feature space in any language. La Cava et al. find generators harder to separate in English [3]; with different models and features we see the opposite on centroid distance, driven by one model, and neither result supports a general convergence claim.

5. Discussion

Interpretable authorship attribution does not require English: the three Class-4 languages average .627 against .675 for the four Class-5 languages, a gap inside the ± 9 -point intervals, and a classifier trained in one language transfers to the others after unlabelled standardization. The choice of representation depends on where the classifier will be used: within one language, with labelled data in that language, character n -grams are the better tool by 3–22 points; across scripts under equal adaptation the features transfer better, and where script and vocabulary are shared, n -grams keep their edge. For this genre and model snapshot, machine translation does not remove the signal, and the surviving features can be named; translation combined with a paraphrase attack is untested, although paraphrase attacks leave a model’s style largely intact in English [13].

Dual use. Nothing in the features is specific to machine-generated text, and human authorship attribution already uses UD-derived features [11], so the release makes public a method rather than a new capability; the translation result is a caution for pseudonymous writers. We release model-generated text and results only.

Limitations. (i) One genre, persuasive essays with three fixed claims and no lists, so paragraph count and length partly record how each model lays out an argument under instruction; both translators also keep paragraph boundaries, so these features survive translation partly by construction. (ii) Length is entangled with style, and the transfer, convergence and translation analyses are not length-controlled. (iii) The resource ranking is coarse, and the RQ2 interval admits a 17-point decline. (iv) Eight of the 21 features depend on Stanza parses or sentence splitting, and parser accuracy varies by language without following the resource order (Hindi and Russian are parsed most accurately, Turkish and Chinese least [18]); four of the five top pooled features are parser-free. (v) The transfer and translation tests share the same 12 prompts across languages, and two generations per cell give intervals of about ± 9 points. (vi) Prompt fidelity was checked by a model, not by native speakers. (vii) All five subjects are closed APIs served in September 2026; open-weight models, adversarial rewriting and a fine-tuned multilingual encoder baseline [4] were not tested. (viii) Judges ran without extended reasoning.

6. Conclusion

Across seven languages, 21 interpretable features with one computational definition in every language attribute persuasive essays to one of five frontier LLMs at .59–.72 accuracy, 3–22 points below character n -grams within a language and .49 against .41 when transferred across languages under the same unlabelled adaptation. Most of the signal sits in a handful of nameable features that differ by language, led in the pooled ranking by paragraph count and paragraph length, which pass through machine translation almost unchanged, so non-English machine text can be attributed with a classifier whose decisions can be read.

Future work: other genres, human-verified prompts, cross-language leave-one-prompt-out transfer, adapted n -grams on translations and a multilingual encoder baseline.

Acknowledgment

Claude Code (Anthropic) assisted in writing the data-collection, feature-extraction and analysis pipeline, and Claude assisted in drafting and editing text. The authors reviewed all AI-assisted code and text and are responsible for the methods, results and conclusions. Code, prompts, responses, translations, judge calls, features, the analysis plan and result tables: <https://github.com/adrian-erlikhman/LangLLM>.

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